

Prashant Piprotar

Sr. Software Engineer

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Profile

Full-Stack Software Engineer with an M.Tech in Computer Science from IIT Kanpur and 5+ years of experience crafting enterprise clean-tech applications. Specialized in backend and full-stack development for Enphase Energy's Solargraf platform, architecting resilient public APIs and automated financial workflow tools (Python/Node.js/React). Blends High-Performance Computing principles with modern web technologies to build low-latency, data-intensive solar management platforms.

Work Experience

- Sr. Software Engineer, Enphase Energy** 08/2021 – Present
Bengaluru
- Revamped the Enlighten Manager dashboard, enabling installers to manage solar systems more effectively, which led to improved operational performance.
 - Developed the BOM Calculator, enabling installers to generate Bills of Materials, which improved customer satisfaction by 5% through real-time pricing integration within 3 months.
 - Led the development of a new user interface for Proposal, enhancing user experience and engagement within 5 months through user feedback integration.
 - Developed the Public API for Solargraf, enhancing external developer access and documentation, which improved integration efficiency within 4 months.
- Intern, Centre for Development of Advanced Computing, R&D** 01/2018 – 04/2018
Pune
- Developed an open-source Network Emulator and IDE, streamlining SDN application development, which enhanced developer productivity within 4 months.
 - Developed a CEPH storage system for the SDN Online Lab, enhancing data accessibility and reliability within 4 months through robust architecture design.

Education

- M. Tech. in Computer Science & Engg., IIT Kanpur** 07/2019 – 06/2021
GPA: 8.6/10
- B. Tech. in Computer Engineering, CSPIT, CHARUSAT** 07/2014 – 05/2018
GPA: 9.04/10
- XII(C.B.S.E.), K.V.No.2, Jamnagar** Jamnagar
GPA: 81%
- X(C.B.S.E.), K.V.No.2, Jamnagar** Jamnagar
GPA: 10/10

Skills

Programming Languages

C# C++ Java JavaScript (ES6+) Python Scala

Backend & APIs

Microservices Node.js RESTful APIs SpringBoot

DevOps & Cloud

AWS Azure Docker GCP Git (CI/CD workflows)

Frontend

MaterialUI React.js Redux TypeScript (migration experience)

Databases

MongoDB (design, optimization, integration) MySQL

Parallel & Distributed Computing

MPI Spark, OpenMPI

Projects

Evaluation of Spark and MPI for Scientific Data Analysis, IIT Kanpur Thesis This thesis compares traditional MPI C++ with Apache Spark for scientific data analysis. Experiments on CPU, memory, and I/O intensive applications from astrophysics, meteorology, and combustion were conducted on multiple clusters.	08/2020 – 05/2021
CloudMPI  , IIT Kanpur - Designed a cloud-augmented MPI support library to reduce network latency and improve performance of distributed computations in virtualized environments - Implemented optimized communication layers over traditional MPI to handle high-latency cloud infrastructure efficiently	01/2020 – 05/2020
Malware Detection using ML, IIT Kanpur Python, scikit-learn Random forest classification Developed machine learning models for malware detection on the given static and dynamic analysis datasets of the programs.	01/2020 – 05/2020
AetherScan  , Next.js, FastAPI, PyTorch, Three.js, Docker - Developed a GPU-accelerated 3D reconstruction pipeline using Facebook's Fast3R model with CUDA, processing multi-view images into dense point clouds - Engineered real-time WebSocket streaming of point cloud data with raw Float32Array buffers, bypassing React state for high-performance rendering of 1M+ points - Containerized full stack with Docker Compose and NVIDIA Container Toolkit for reproducible GPU-enabled deployment	2026
Real-Time Object Detection  , Python, PyTorch, CUDA, Streamlit, OpenCV - Developed a GPU-accelerated real-time object detection system using YOLO26 with CUDA for low-latency video inference - Engineered an interactive Streamlit dashboard featuring dynamic FPS monitoring, custom glassmorphic styling, and threshold sliders	

Certifications

MTA: Windows Server Administration Fundamentals-Certified 2017 Microsoft	Mobile Communication BSNL RTTC Ahmedabad
Cisco Certified Network Associate: Routing and Switching Cisco 07/2015 - 04/2017	

Awards & Scholarships

All India Rank 422, GATE	01/2019
Certificate of Merit for class X, CBSE	